Fpt University

TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 1 – Project Introduction

– Hanoi, Sep 2026 –

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Change Log

Date	A* M, D	In charge	Change Description
20/9/2026	A	Nguyen Hoang Thang	Add I.Overview
20/9/2026	A	Nguyen Hoang Thang	Add II.Product Background
20/9/2026	A	Nguyen Thanh Luan	Add III.Existing Solutions
20/9/2026	A	Nguyen Viet Hoang	Add IV. Proposed Solution
21/9/2026	M	Nguyen Thanh Luan	Add detailed information to IV. Proposed Solution
21/9/2026	A	Nguyen Chi Tung Duong	Add V.Project Scope & Limitations

*A - Added M - Modified D - Deleted

Definition and Acronyms

Acronym	Definition
AI	Artificial Intelligence
GPS	Global Positioning System
QR	Quick Response (Code)
API	Application Programming Interface
GAP	Identified gap in existing solutions
FE	Feature
LI	Limitation / Exclusion
EPR	Extended Producer Responsibility
B2C	Business-to-Consumer
B2B	Business-to-Business
MVP	Minimum Viable Product
UI/UX	User Interface / User Experience

I. Project Overview

1. Project Information

- English name: ***Intelligent Recyclable Waste Classification and Trade Matching System.***
- Vietnamese name: ***Hệ thống hỗ trợ phân loại và kết nối thu mua rác thải tái chế thông minh.***
- Project code: ***EcoLink***
- Group name: ***SEP490-G141***
- Software type: ***Web Application***

2. Project Purpose

The primary purpose of this project is to build a system that helps users easily identify types of waste through captured or uploaded images.

The system not only recognizes the type of waste but also provides step-by-step sorting instructions so users know exactly how to handle it.

Additionally, the project aims to connect users with smart recycling purchasing facilities to promote and support green living habits.

3. Project Stakeholders

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II. Product Background

Based on the practical issues outlined in the project's research, sorting waste at the source currently faces a massive barrier stemming from a very simple reality: users are often unsure where a specific type of waste should be thrown.

Imagine a daily scenario: You have just finished a meal in a styrofoam box or drank a bottle of water. Standing in front of trash bins labeled "Organic", "Recyclable", and "Other", you start to wonder. Should a styrofoam box with food residue go into the recycling bin or the other bin? For a plastic bottle, which bin does it go to? Or for items like old batteries, how can they be handled since they cannot be thrown out with regular household waste?

Because there are no ready answers at that moment, users still have to make their own decisions on the spot. Even with a good intention to protect the environment, this confusion leads to improper disposal, causing recyclable materials to be ruined or sent to landfills.

The core problem here is two-fold. First, when looking at an object, users simply do not know what category it belongs to (e.g., whether it is plastic, paper, metal, or other waste). Second, even if they know what the item is (for example, a plastic bottle), they do not know the proper handling process, such as whether it needs to be washed, if it is recyclable, or which specific bin it belongs to.

This is the exact reason the EcoLink system was created. Instead of forcing users to memorize complicated waste sorting rules, the system provides an extremely simple workflow: the user takes or uploads a photo of the waste, the AI recognizes the item, and the system looks up the sorting rules to provide a clear instruction. For example, if a user snaps a picture of a water bottle, the system will immediately identify it as a plastic bottle and provide visual handling instructions, such as: "1. Empty liquid, 2. Rinse if necessary, 3. Put it into the recycling bin". By providing guidance right at the moment of disposal, the product turns a confusing daily chore into a confident and eco-friendly habit.

III. Existing Solutions

Four categories of solutions currently address parts of the recyclable waste problem. The analysis below evaluates each against the specific context of EcoLink: a Vietnamese platform that must (1) identify waste from a photo, (2) deliver localised sorting instructions, and (3) connect users with trade partners who will actually purchase the material.

1. Recycle Coach

Recycle Coach is a waste-management mobile application developed by Recycle Coach Inc. (Canada). It is deployed in partnership with hundreds of municipalities across North America and Australia, allowing each city to configure its own rules and collection schedules directly inside the platform.

Website / Link	https://www.recyclecoach.com/
Target Users	Urban and suburban residents in North America, Australia; municipalities seeking a citizen-facing waste tool
Core Features	"What Goes Where?" text-based item search; AI-assisted camera scan (limited rollout); personalised curbside collection calendar; automated pickup reminders; drop-off depot locator; in-app issue reporting to local authorities
Strengths	Comprehensive rule database tied to real municipal policies; official integration with local governments ensures accuracy; calendar and reminder features drive repeat engagement
Weaknesses	Camera recognition is optional and inconsistently available across municipalities; the platform has no trade or purchase mechanism — it only tells users *where* to drop items, not *who will buy* them; service coverage is concentrated in North America and Australia
GAP for this project	Recycle Coach is built around the assumption that a municipality will manage the platform and keep its database current. In Vietnam, no such municipality-level

	digital infrastructure exists. Furthermore, the app has no mechanism for users to connect with private scrap-buying partners (thu mua) or to complete a buy-sell transaction. A Vietnamese user who correctly identifies a plastic bottle using Recycle Coach is still left with no actionable next step for monetising or handing over that material.

2. iRecycle

iRecycle, developed by Earth911 (USA), was one of the largest recycling-locator tools in North America, providing access to a database of over 100,000 recycling and disposal facilities across hundreds of material categories.

Website / Link	https://search.earth911.com/
Target Users	US-based residents seeking drop-off or recycling locations for specific materials
Core Features	Location and ZIP-code-based search for nearby recycling facilities; category browsing (plastics, metals, electronics, batteries, etc.); facility details including contact, hours, and accepted materials
Strengths	Extensive material database covering rare items such as motor oil, paint, and e-waste; the most comprehensive recycling-location tool available for the US market
Weaknesses	Geographic coverage is almost exclusively the United States; no AI image recognition — users must already know what material they have; no buying/selling functionality; mobile app has been discontinued, leaving only a web tool; no personalisation or gamification
GAP for this project	iRecycle is a static directory, not an intelligent assistant. It requires users to know the exact material name before searching — precisely the knowledge gap that EcoLink targets. Beyond geography (the tool has zero Vietnam coverage), iRecycle cannot help a user who holds an ambiguous object and wonders whether it is recyclable. It also has no mechanism for a private recycling trader to publish their buying price, availability, or accepted categories, which is the second core value EcoLink must deliver.

3. Litterati

Litterati is a US-based citizen-science application that uses AI (branded as LitterAI) to classify litter items photographed during community clean-up activities. Its primary goal is to generate structured environmental datasets that NGOs and policymakers can use to drive systemic change.

Website / Link	https://www.litterati.org/
Target Users	Environmental volunteers, NGOs, citizen-science communities, schools, corporate CSR programs
Core Features	Photo capture with AI-assisted tagging (Object–Material–Brand taxonomy); GPS geotagging and timestamping; community litter maps; brand-pollution analytics; data export for policymakers
Strengths	The only solution reviewed that applies AI image analysis directly to waste photos; large-scale dataset validated by National Science Foundation funding; effective at creating accountability for brand-level plastic pollution
Weaknesses	The AI classifies litter for *research and accountability purposes*, not for giving individual users sorting instructions; the app is designed for outdoor litter photography, not for household disposal decisions; there is no knowledge base of sorting rules, no handling guidance, and no connection to recycling buyers

GAP for this project	Litterati demonstrates that AI-based waste photo classification is feasible, but it is built entirely for community data collection — not for the moment-of-disposal decision a household user faces. It answers "what brand polluted here?" rather than "what should I do with this item right now?". Litterati does not provide step-by-step handling instructions, has no knowledge base tied to Vietnamese waste policy, and has no feature for connecting users with trade partners. It addresses the analytical layer while leaving the practical user need unsolved.

4. VECA

VECA is a Vietnamese technology startup (founded 2021) that built an "Uber-like" mobile platform to connect households wanting to sell recyclable scrap with licensed scrap collectors and recycling facilities. It was the closest local equivalent to EcoLink's trade-matching feature.

Website / Link	https://veca.app/doanh-nghiep/phe-lieu-cong-nghiep
Target Users	Urban Vietnamese households with recyclable scrap to sell; scrap collectors and recycling businesses (B2B focus since late 2024)
Core Features	On-demand home scrap-collection booking; transparent price list for common materials (plastic, paper, metal); digital management tools for scrap yards (VECA Vựa Số); e-wallet payment integration; corporate partnership program (EPR compliance)
Strengths	First mover in digitising Vietnam's traditionally informal ve-chai (scrap collector) sector; transparent, standardised pricing reduces exploitation of both sellers and collectors; praised for intuitive UI despite a technically diverse user base
Weaknesses	B2C home-collection service was officially discontinued on 1 October 2024 due to unsustainable unit economics (small-volume orders were unprofitable); no AI waste-recognition feature — users must already know what they have and how much of it; no sorting guidance or knowledge base; pivoted fully to B2B scrap-yard management tools, leaving individual household users without a solution
GAP for this project	VECA proved that Vietnamese users are willing to engage with a digital recycling platform when price transparency and convenience are offered. However, the discontinuation of its B2C service exposed the unsolved problem: individual users produce small, mixed, and often unidentified waste that does not fit neatly into a booking-based model. EcoLink addresses the layer that VECA skipped — helping users first identify and classify what they have — before connecting them to a trade partner. The addition of AI recognition, a sorting knowledge base, and a gamified engagement loop is precisely what is needed to make the connection sustainable at the individual-user level.

5. Section Summary

Each of the solutions reviewed addresses at most one of EcoLink's three core requirements. Recycle Coach and iRecycle provide sorting guidance but are unavailable in Vietnam, rely on text-based input (requiring users to already know what they hold), and have no mechanism for completing a trade transaction. Litterati introduces AI image classification but uses it exclusively for environmental research, providing no actionable disposal instructions and no buyer connection. VECA demonstrated real demand for a Vietnamese recycling trade platform but folded its B2C service because it never solved the upstream problem: users did not know what waste they had or how to handle it. No existing solution combines AI-powered visual identification, localised Vietnamese sorting guidance, and a verified trade-partner marketplace into a single workflow — which is the specific gap EcoLink is designed to fill.

These unresolved deficiencies define three distinct gaps that EcoLink must address: GAP-01 (Waste Identification Gap) — no solution provides AI-powered photo-based identification accessible to Vietnamese users; GAP-02 (Sorting & Handling Guidance Gap) — no solution delivers localised, rule-based disposal instructions at the moment of need; and GAP-03 (Recycling Trade Connectivity Gap) — no solution connects individual households with verified local buyers through a transparent transaction workflow.

IV. Proposed Solution

The platform is an intelligent waste sorting and guidance solution designed to bridge the critical gap between environmental intent and daily execution. As identified in the gap analysis above, no existing solution simultaneously addresses waste identification and localized sorting guidance in the Vietnamese context: international tools such as Recycle Coach and iRecycle lack local coverage, and Litterati's AI serves research rather than individual disposal decisions. Instead of leaving consumers to second-guess static bin labels or misclassify ambiguous household items, it immediately clarifies sorting decisions by identifying waste items from everyday snapshots and providing direct disposal instructions.

Everyday consumers and households benefit directly from this solution through the complete removal of decision fatigue at the bin, transforming an inconvenient chore into a frictionless habit. Additionally, users and recyclers can utilize VietQR for internal system payments, such as removing advertisements or purchasing verified badges.

A photo-based identification approach is the right strategy today because municipal mandates for waste segregation are tightening, yet compliance consistently falters at the moment of disposal. Meeting users precisely where their confusion begins — looking at an object in hand — lowers the cognitive threshold for proper sorting, promoting long-term sustainable recycling behaviour.

V. Project Scope & Limitations

GAP-01: Waste Identification Gap

GAP-02: Sorting & Handling Guidance Gap

GAP-03: Recycler Discovery Gap

1. Major Features

ID	Feature Name	Description	Addressed Gaps
FE-01	AI-Based Waste Identification & Classification	Users capture or upload an image of a waste item, and the trained AI model analyses the image to identify its supported recyclable waste type or material category. The classification result serves as the basis for subsequent sorting guidance and recycling activities.	GAP-01
FE-02	Localized Waste Sorting & Handling Guidance	Based on the identified waste category, the system provides step-by-step instructions explaining whether the item is recyclable and how it should be prepared, cleaned, separated, and sorted according to the configured waste-handling rules.	GAP-02
FE-03	Waste Knowledge Base Management	The system maintains structured information about supported waste categories and their corresponding sorting and handling instructions. Authorized administrators can maintain this information so that guidance can be updated without retraining the AI model.	GAP-01, GAP-02
FE-04	Recyclable Waste Listing Management	Users create listings for recyclable waste they want to sell or hand over, including waste category, images, estimated quantity, location, and other relevant information.	GAP-03

		Classification results can be reused when creating a listing to reduce manual input.	
FE-05	Recycler Profile & Purchasing Information Management	Scrap collectors and recycling facilities maintain profiles containing their location, accepted waste categories, purchasing prices or conditions, and availability. This information allows users and the matching mechanism to determine whether a recycler is suitable for a particular recyclable material.	GAP-03
FE-06	Location-Based Recycler Discovery	Users can search for scrap collectors or recycling facilities that accept a specific recyclable material and are geographically relevant to their location. Location and distance information are used to improve the relevance of recycler discovery.	GAP-03
FE-07	Intelligent Trade Matching	The system matches recyclable-waste listings with suitable recyclers based on factors such as waste category, estimated quantity, geographic location, purchasing conditions, and recycler availability. The results help users identify appropriate recycling trade partners.	GAP-03
FE-08	Internal System Payments (VietQR Integration)	Users and recyclers can make internal system service payments using VietQR, such as paying to remove advertisements or purchasing verified badges to enhance credibility.	GAP-03
FE-09	In-System Notifications	The system provides in-system notifications for important user events such as service payment confirmations, account status updates, or new features.	GAP-03
FE-10	Account & System Data Management	Administrators manage user and recycler accounts, service payments, waste categories, sorting rules, and reference data required to support classification, guidance, and recycler discovery workflows.	GAP-01, GAP-02, GAP-03

2. Limitations & Exclusions

ID	Excluded Item	Reason
LI-01	Native Mobile Application	The current version is developed as a web application. Dedicated Android and iOS applications require additional development, testing, and maintenance effort and are outside the scope of the initial capstone release.
LI-02	Automated Physical Waste S orting	EcoLink provides software-based waste identification and sorting guidance only. Integration with robotic sorting systems, smart bins, conveyor systems, or other physical waste-processing equipment is not included in this version.
LI-03	Recognition of Unsupported Waste Categories	The AI model is limited to the recyclable waste categories represented in the project's training dataset. Objects or waste types outside the supported classes are not guaranteed to be identified correctly.
LI-04	Automatic Waste Weight and Quality Verification	The web application cannot physically verify the exact weight, cleanliness, contamination level, or material quality of recyclable waste. Such information must be provided by the user and may be confirmed by the recycler during the actual handover.
LI-05	Waste Transportation & Fleet Management	EcoLink connects users with scrap collectors and recycling facilities but does not directly operate transportation services or manage vehicles, drivers, fleet schedules, or physical waste collection operations.
LI-06	Nationwide Recycler Coverage Guarantee	Recycler discovery and matching depend on the collectors and recycling facilities available in the EcoLink database. The system therefore cannot guarantee complete coverage of all recycling purchasers throughout Vietnam.
LI-07	Guaranteed AI Classification Accuracy	Classification accuracy depends on factors such as the training dataset, image quality, lighting conditions, viewing angle, background, and

		similarity between waste categories. AI results are intended to assist users and cannot guarantee correct identification in every case.
LI-08	Real-Time Recycling Market Price Prediction	EcoLink may present purchasing prices or conditions provided by recyclers, but it does not operate as a commodity exchange and does not predict or guarantee real-time market prices for recyclable materials.
LI-09	In-App Transaction Processing	EcoLink provides recycler discovery and internal system payment integrations (VietQR for ad removal/verified badges), but does not process or mediate physical scrap trade transactions or payments between users and recyclers.